

Lecture Nine

1 Application of Differentiation

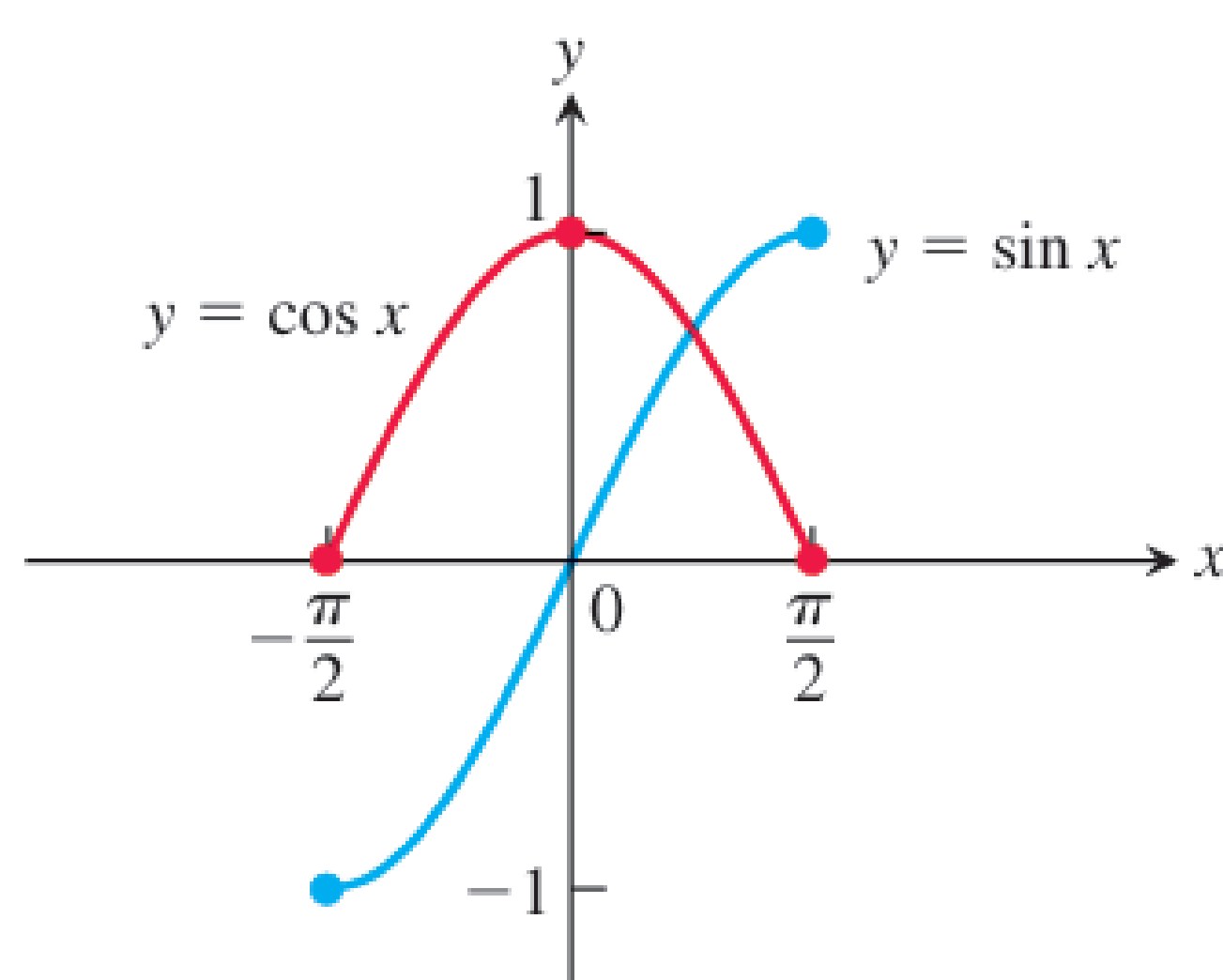
1.1 Extreme Values of functions on Closed Intervals

In this section we will show that how to locate and identify extreme (max or min) values of a function from its derivative. Once we can do this, we can solve a variety of optimization problems. The domain of the functions we consider are intervals or union of separate intervals.

Definition 1.1.1 Let f be a function with domain D . Then f has an **absolute maximum value** on D at a point c if $f(x) \leq f(c)$, $\forall x \in D$ and an **absolute minimum value** on D at a point c if $f(x) \geq f(c)$, $\forall x \in D$.

Max and Min values are called **extreme values** of the function f . Absolute maxima or minima are referred to **global** maxima or minima.

Example 1.1.1 On the closed interval $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ the function $f(x) = \cos x$ takes on an absolute maximum value of 1 (once) and an absolute minimum value of 0 (twice) on the same interval, the function $g(x) = \sin x$ takes on a maximum value of 1 and a minimum value of -1 .



Example 1.1.2 The absolute extrema of the following functions on their domains can be seen in the following figure. Each function has the same defining equation, $y = x^2$, but the domains vary.

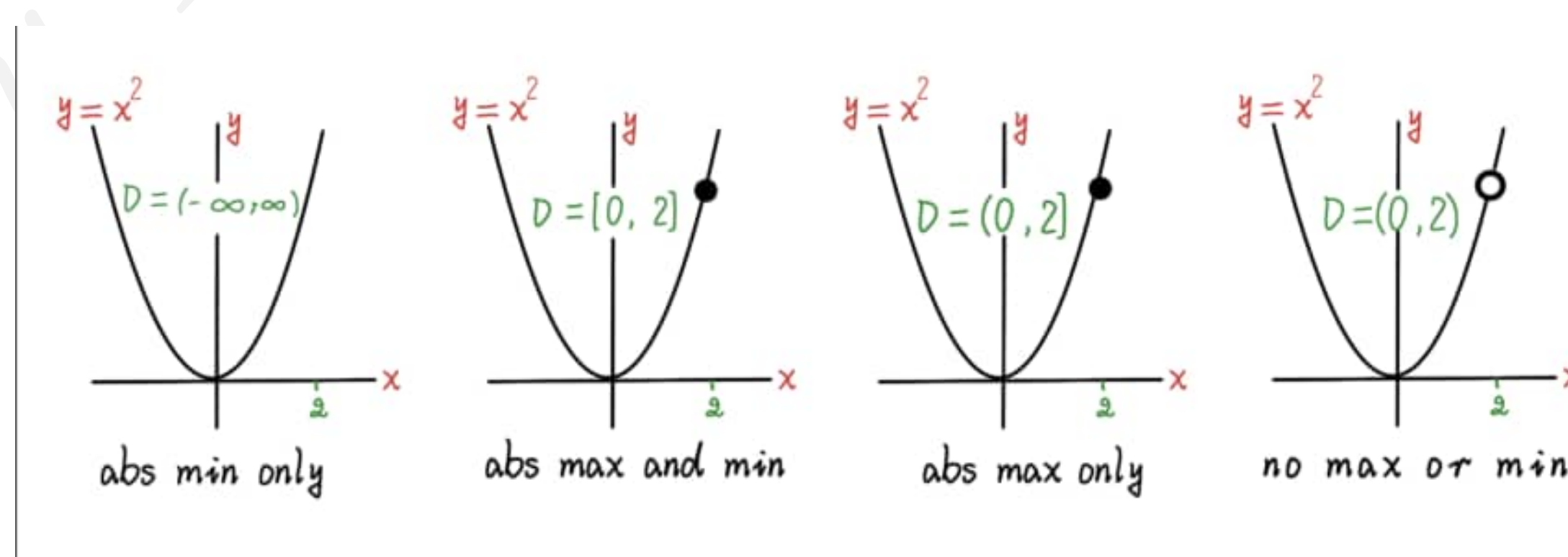
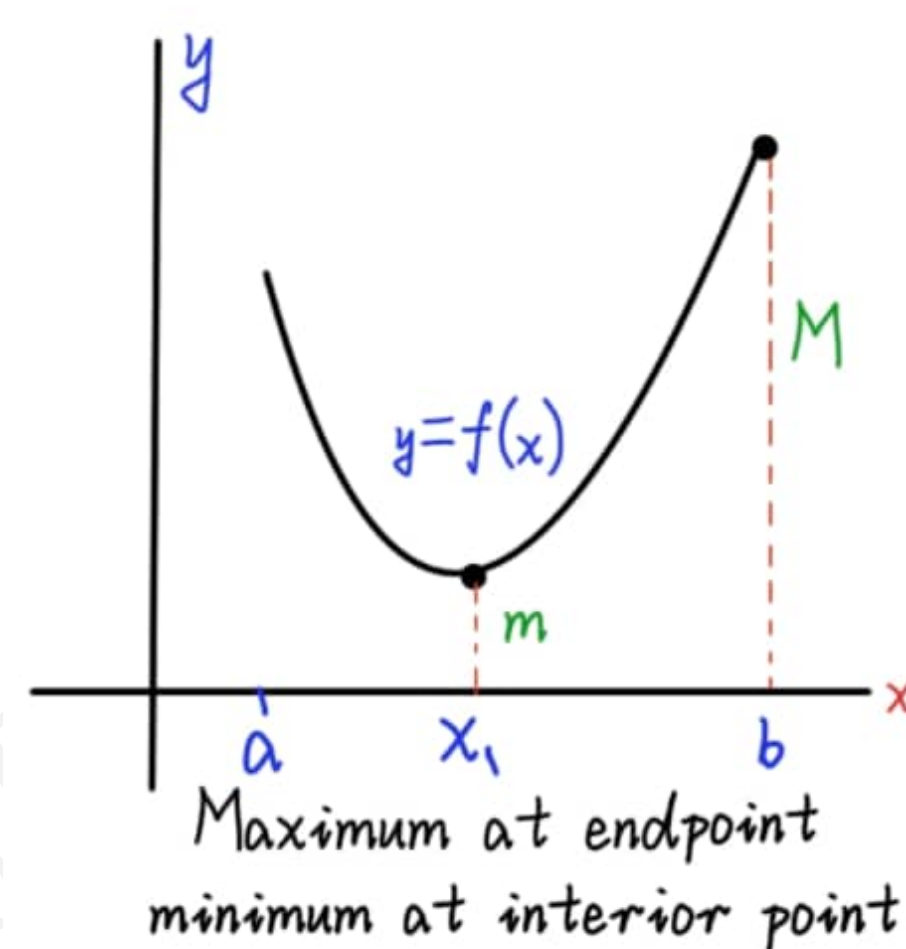
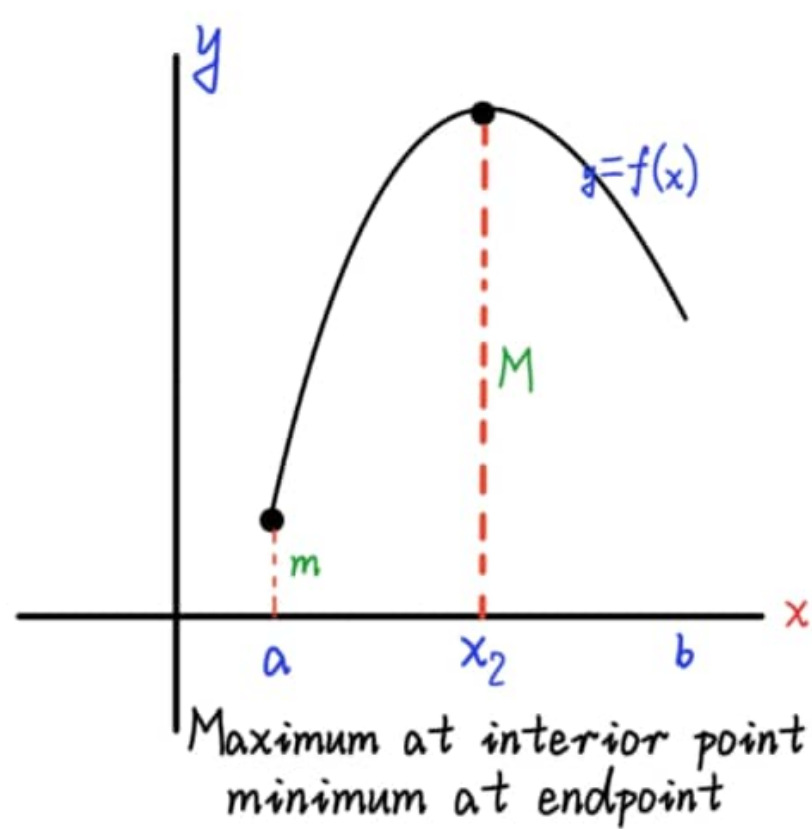
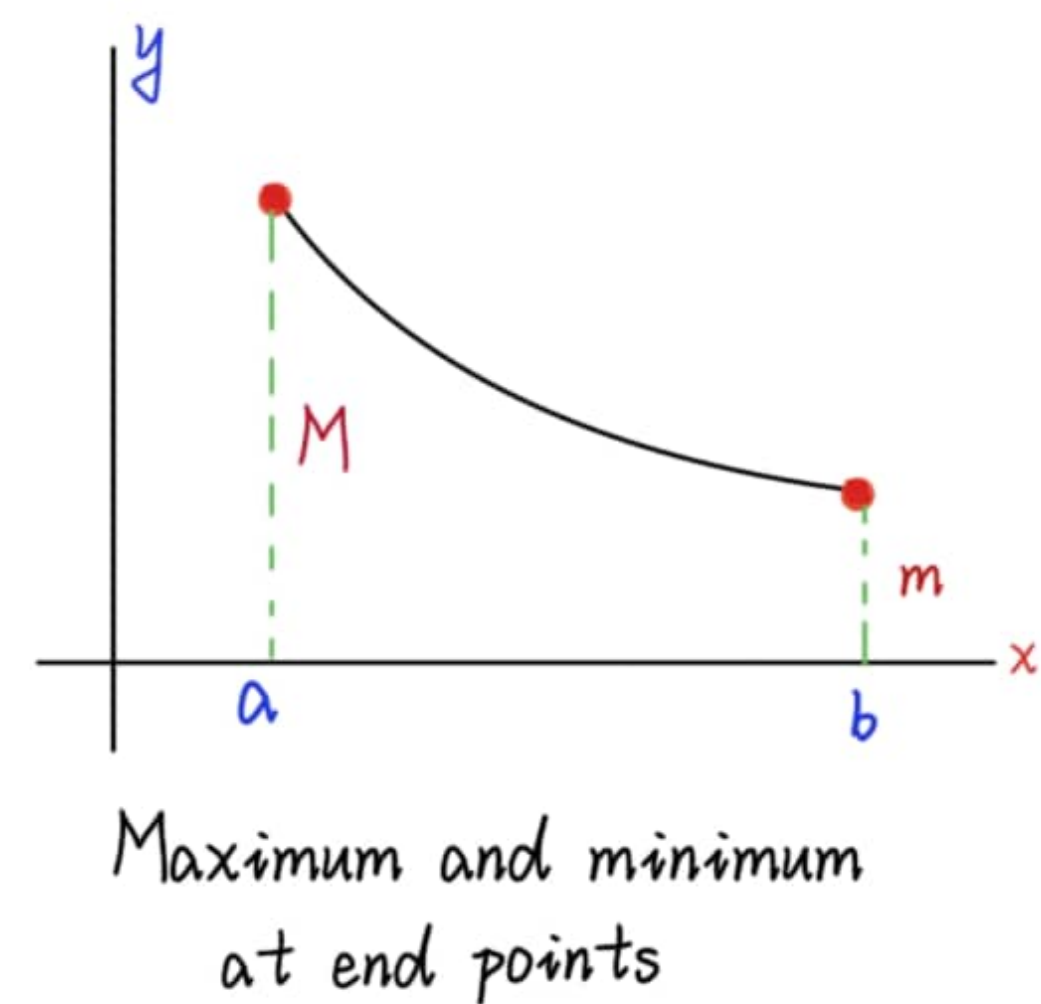
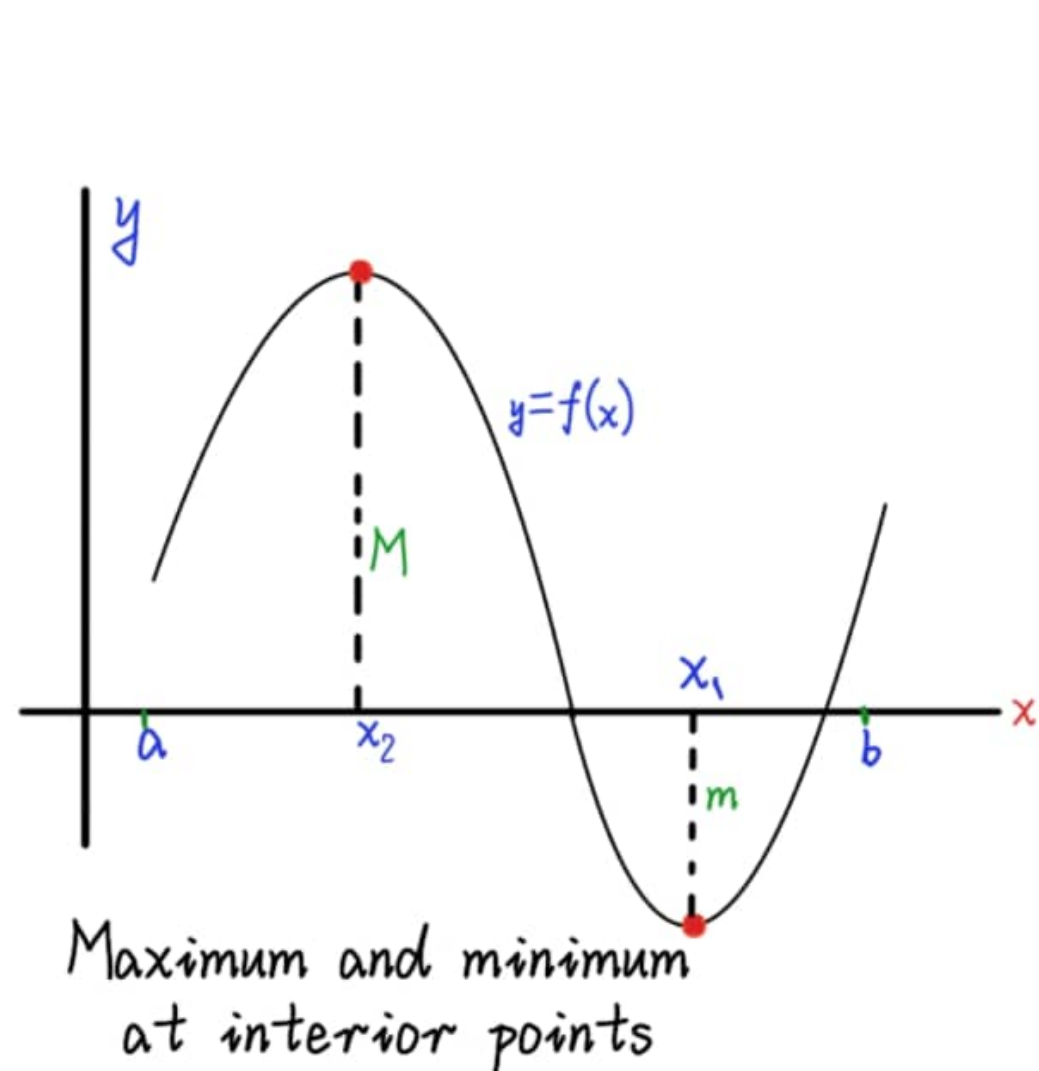


Figure 1: Graphs for example 1.1.2

Function Rule	D	Absolute extrema on D
$y = x^2$	$(-\infty, \infty)$	No abs max and abs min at $x = 0$
$y = x^2$	$[0, 2]$	abs max of 4 at $x = 2$ and abs min of 0 at $x = 0$
$y = x^2$	$(0, 2]$	Abs max of 4 at $x = 2$ and no abs min
$y = x^2$	$(0, 2)$	No abs extrema

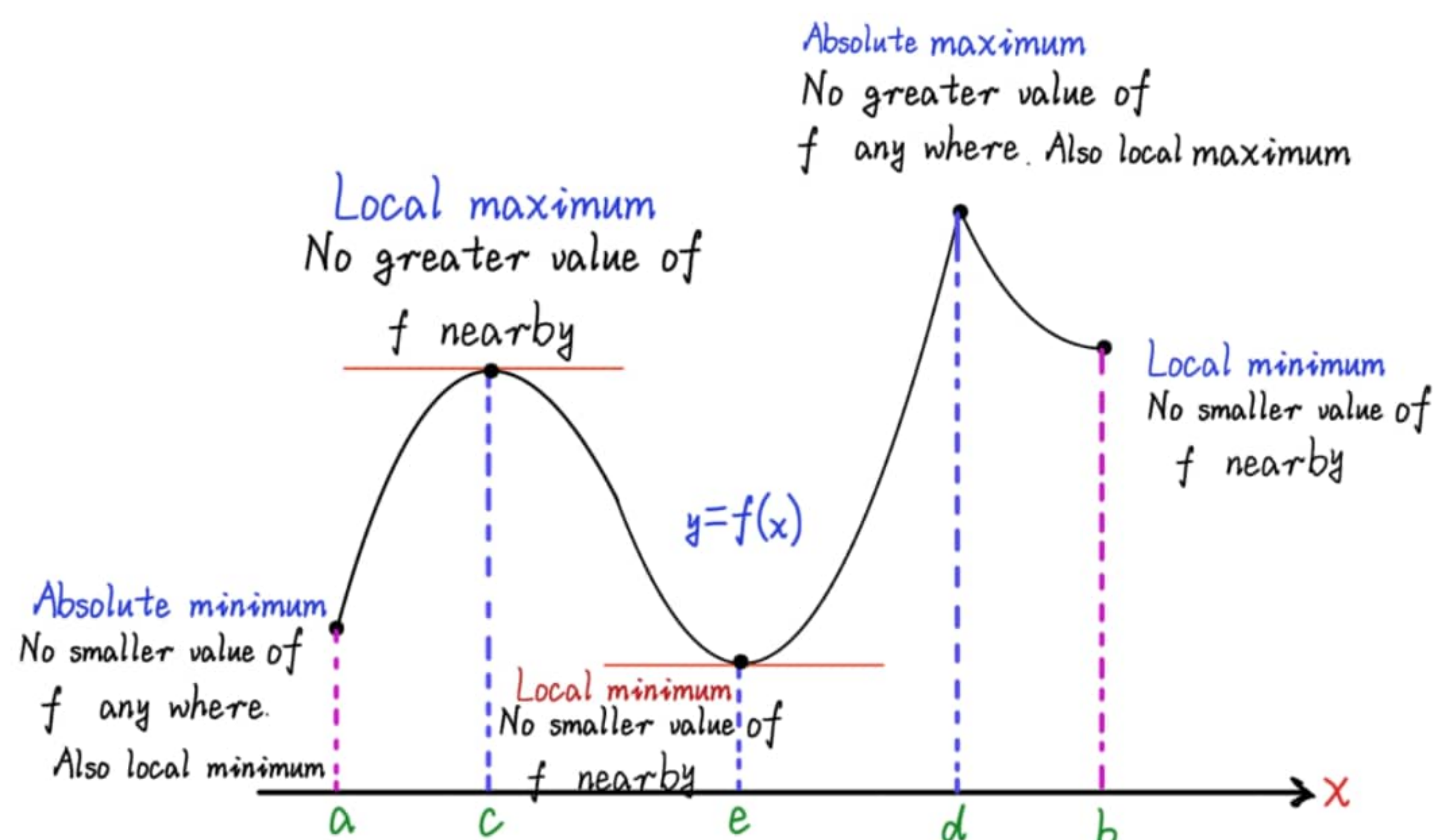
Theorem 1.1.1 The Extreme Value Theorem:

If f is continuous on a closed interval $[a, b]$, then f attains both an absolute maximum value M and an absolute minimum value m in $[a, b]$. That is, there are numbers x_1 and x_2 in $[a, b]$ with $f(x_1) = m$, $f(x_2) = M$, and $m \leq f(x) \leq M$ for every other x in $[a, b]$.

**1.1.1 Local (Relative) Extreme Values**

Definition 1.1.2 A function f has a **local maximum** value at a point c within its domain D if $f(x) \leq f(c)$, $\forall x \in D$ lying in some open interval containing c .

A function f has a **local minimum** value at a point c within its domain D if $f(x) \geq f(c)$, $\forall x \in D$ lying in some open interval containing c .



Theorem 1.1.2 The First Derivative Theorem for Local Extreme Values:

If f has local maximum or minimum value at an interior point c of its domain, and if f' is defined at c , then

$$f'(c) = 0$$

Definition 1.1.3 An interior point of the domain of a function f where f' is zero or undefined is a **critical point** of f .

Finding The Absolute Extrema of a Continuous Function:

1. Find all critical points of f on the interval.
2. Evaluate f at all critical points and endpoints.
3. Take the largest and smallest of these values.

Example 1.1.3 Find the absolute maximum and minimum values of $f(x) = x^2$ on $[-2, 1]$.

Sol

The function is differentiable over its entire domain, so the only critical point occurs where $f'(x) = 2x = 0$ when $x = 0$.

We need to check the function's values at $x = 0$ and at the endpoints $x = -2$ and $x = 1$.

$$\begin{array}{ll} \text{Critical point value} & f(0) = 0 \\ \text{Endpoints values} & f(-2) = 4 \text{ and } f(1) = 1 \end{array}$$

Therefore the function has an absolute maximum value of 4 at $x = -2$ and absolute minimum value of 0 at $x = 0$.

Example 1.1.4 Find the absolute maximum and minimum values of $g(t) = 8t - t^4$ on $[-2, 1]$.

Sol

The function is differentiable on its entire domain, so the only critical points occur where $g'(t) = 0$. Solving this equation gives:

$$8 - 4t^3 = 0 \Rightarrow t = \sqrt[3]{2} > 1$$

a point not in the given domain. The function's absolute extrema therefore occur at the endpoints

$$\begin{array}{ll} g(-2) = -32 & \text{Abs min} \\ g(1) = 7 & \text{Abs max} \end{array}$$

Example 1.1.5 Find the absolute maximum and minimum values of $f(x) = x^{\frac{3}{2}}$ on the interval $[-2, 3]$

Sol

Exercise 1.1.1 Find the Abs min and max for the following functions.

1. $f(x) = x^2 - 1$, $-1 \leq x \leq 2$
2. $f(x) = 4 - x^3$, $-2 \leq x \leq 1$
3. $f(x) = \sqrt{4 - x^2}$, $-2 \leq x \leq 1$
4. $g(x) = \sqrt[3]{x}$, $-1 \leq x \leq 8$
5. $f(x) = \frac{2}{3}x - 5$, $-2 \leq x \leq 3$

1.2 Monotonic Functions and the First Derivative Test

Definition 1.2.4 A function f is **increasing** on an interval if for any two numbers x_1 and x_2 in the interval

$$x_1 < x_2 \Rightarrow f(x_1) < f(x_2)$$

A function f is **decreasing** on an interval if for any two numbers x_1 and x_2 in the interval

$$x_1 < x_2 \Rightarrow f(x_1) > f(x_2)$$

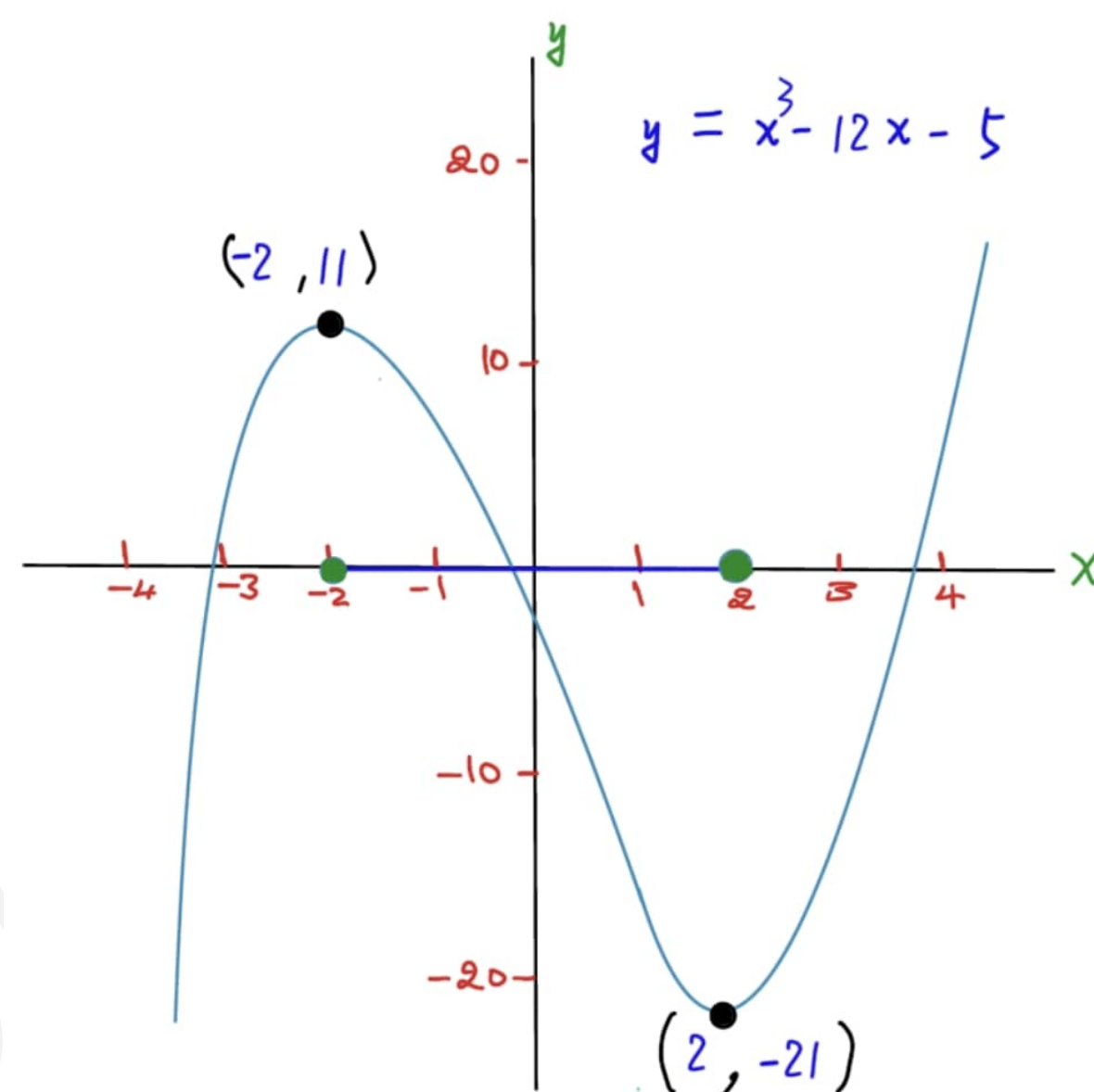
If a function $f(x)$ is increasing or decreasing then it is **monotonic**.

Theorem 1.2.3 Suppose that f is continuous on $[a, b]$ and differentiable on (a, b)

1. If $f'(x) > 0$ at each point $x \in (a, b)$, then f is increasing on $[a, b]$.
2. If $f'(x) < 0$ at each point $x \in (a, b)$, then f is decreasing on $[a, b]$.

Example 1.2.6 Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the open intervals on which f is increasing and on which f is decreasing.

Sol



Clearly f is everywhere continuous and differentiable. The first derivative:

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4) = 3(x + 2)(x - 2)$$

is zero at $x = -2$ and $x = 2$. These critical points subdivide domain of f to create non overlapping open intervals $(-\infty, -2)$, $(-2, 2)$ and $(2, \infty)$ on which f' is either positive or negative.

We determine the sign of f' by evaluating f' at the critical point $\pm\epsilon$.

1. When $x < -2 \Rightarrow x \in (-\infty, -2)$

$$f'(-2 - \epsilon) = 3(-2 - \epsilon + 2)(-2 - \epsilon - 2) = 3(-\epsilon)(-4 - \epsilon) = 3(\epsilon)(4 + \epsilon) > 0$$

Hence the function is increasing on the interval $(-\infty, -2)$

2. When $x \in (-2, 2)$

$$(i) f'(-2 + \epsilon) = 3(-2 + \epsilon + 2)(-2 + \epsilon - 2) = 3(\epsilon)(\epsilon - 4) < 0$$

$$(ii) \quad f'(2 - \epsilon) = 3(2 - \epsilon + 2)(2 - \epsilon - 2) = 3(4 - \epsilon)(-\epsilon) < 0.$$

Therefore the function decreasing on $(-2, 2)$

3. When $x > 2 \Rightarrow x \in (2, \infty)$

$$f'(2 + \epsilon) = 3(2 + \epsilon + 2)(2 + \epsilon - 2) = 3(4 + \epsilon)(\epsilon) > 0$$

Therefore the function is increasing on $(2, \infty)$.

Example 1.2.7 Find the critical points of $f(x) = x^3 - 6x^2$ and identify the open intervals on which f is increasing and on which f is decreasing.

Sol

Firstly we find $\frac{dy}{dx}$.

$$\frac{dy}{dx} = 3x^2 - 12x = 3x(x - 4)$$

The intervals on which the function increasing, $\frac{dy}{dx} = 3x(x - 4) > 0$.

(a) Either $x > 0$ and $x - 4 > 0 \Rightarrow x > 4 \equiv x \in (4, \infty)$

(b) or $x < 0$ and $x - 4 < 0 \Rightarrow x < 0 \equiv x \in (-\infty, 0)$

And the intervals on which the function is decreasing, $\frac{dy}{dx} < 0$

(i) Either $x < 0$ and $x - 4 > 0$

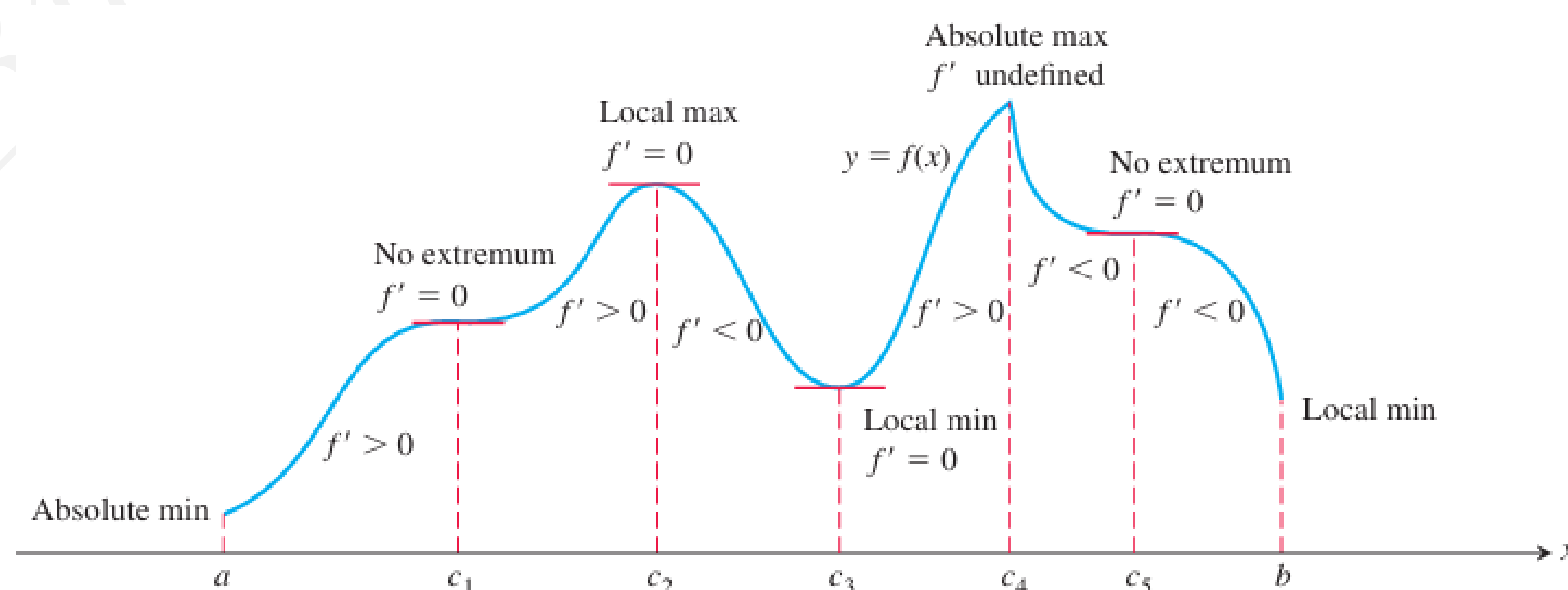
$$\Rightarrow (-\infty, 0) \cap (4, \infty) = \phi$$

(ii) or $x > 0$ and $x - 4 < 0$

$$\Rightarrow x \in (0, \infty) \cap (-\infty, 4) \Rightarrow x \in (0, 4)$$

Therefore the function will be decreasing on the interval $(0, 4)$

1.3 First Derivative Test for Local Extrema



Suppose that c is a critical point of a continuous function f , and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across this interval from left to right.

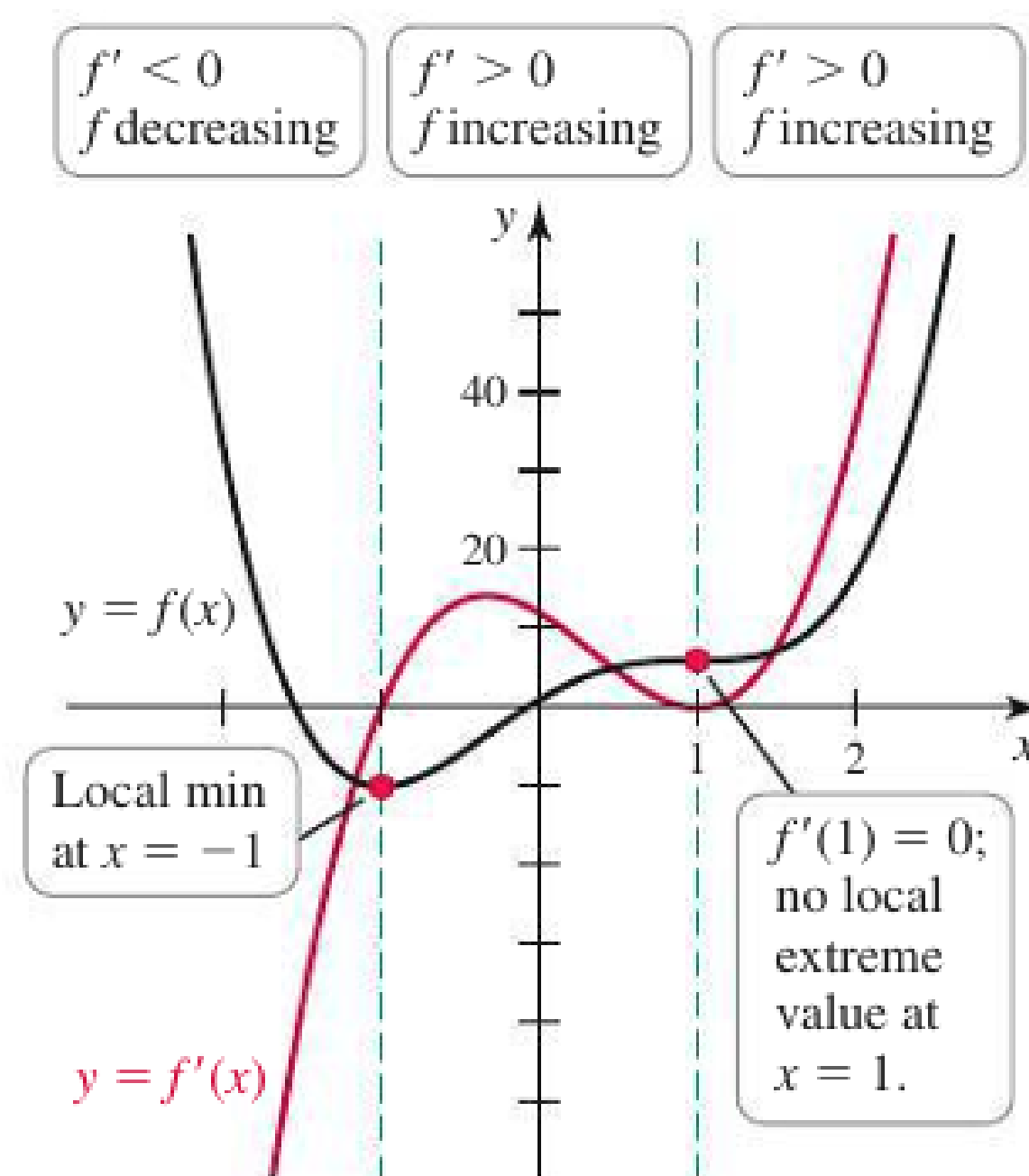
1. If f' changes from $(-)$ to $(+)$ at c , then f has a local minimum at c .

2. If f' changes from $(+)$ to $(-)$, then f has a local maximum at c .
3. If f' does not change sign at c , i.e. f' is positive on both sides or negative on both sides, then f has no local extremum at c .

Example 1.3.8 By using the first derivative test, consider the function $f(x) = 3x^4 - 4x^3 - 6x^2 + 12x + 1$

- (a) Find the intervals on which f is increasing and those on which it is decreasing.
- (b) Identify the local extrema of f .

Sol



Differentiating f , we find that:

$$f'(x) = 12x^3 - 12x^2 - 12x + 12 = 12(x^3 - x^2 - x + 1) = 12(x + 1)(x - 1)^2$$

Solving $f'(x) = 0$ gives the critical points $x = -1$ and $x = 1$. The critical points determine the intervals $(-\infty, -1)$, $(-1, 1)$ and $(1, \infty)$ on which f' does not change sign.

1. When $x < -1$, $x \in (-\infty, -1)$ substitute $-1 - \epsilon$ on $f'(x)$.

$$f'(-1 - \epsilon) = 12(-1 - \epsilon + 1)(-1 - \epsilon - 1)^2 = 12(-\epsilon)(-2 - \epsilon)^2 < 0$$

$\Rightarrow f$ is decreasing on $(-\infty, -1)$

2. When $x \in (-1, 1)$

$$(i) f'(-1 + \epsilon) = 12(-1 + \epsilon + 1)(-1 + \epsilon - 1)^2 = 12(\epsilon)(\epsilon - 2)^2 > 0$$

$$(ii) f'(1 - \epsilon) = 12(1 - \epsilon + 1)(1 - \epsilon - 1)^2 = 12(2 - \epsilon)(-\epsilon)^2 > 0$$

Here $f(x)$ is increasing on $(-1, 1)$

3. When $x \in (1, \infty)$; $x > 1$

$$f'(1 + \epsilon) = 12(1 + \epsilon + 1)(1 + \epsilon - 1)^2 = 12\epsilon^2(2 + \epsilon) > 0$$

$f(x)$ is increasing on $(1, \infty)$.

Now we want to identify the local extrema of f at $x = -1, x = 1$. We will check the changing derivative's sign before and after the critical points.

1. At $x = -1$

$$(i) f'(-1 - \epsilon) = 12(1 - \epsilon + 1)(-1 - \epsilon - 1)^2 = 12(-\epsilon)(-2 - \epsilon)^2 < 0$$

$$(ii) f'(-1 + \epsilon) = 12(-1 + \epsilon + 1)(-1 + \epsilon - 1)^2 = 12(\epsilon)(-2 + \epsilon)^2 > 0$$

From $(-)$ to $(+)$ $\Rightarrow$ local minimum at $x = -1$

2. At $x = 1$

$$(i) f'(1 - \epsilon) = 12(1 - \epsilon + 1)(1 - \epsilon - 1)^2 = 12(2 - \epsilon)(-\epsilon)^2 > 0$$

$$(ii) f'(1 + \epsilon) = 12(1 + \epsilon + 1)(1 + \epsilon - 1)^2 = 12(2 + \epsilon)(\epsilon)^2 > 0$$

Observe that f' is positive on both sides near $x = 1$, so f does not have a local extreme value at $x = 1$.

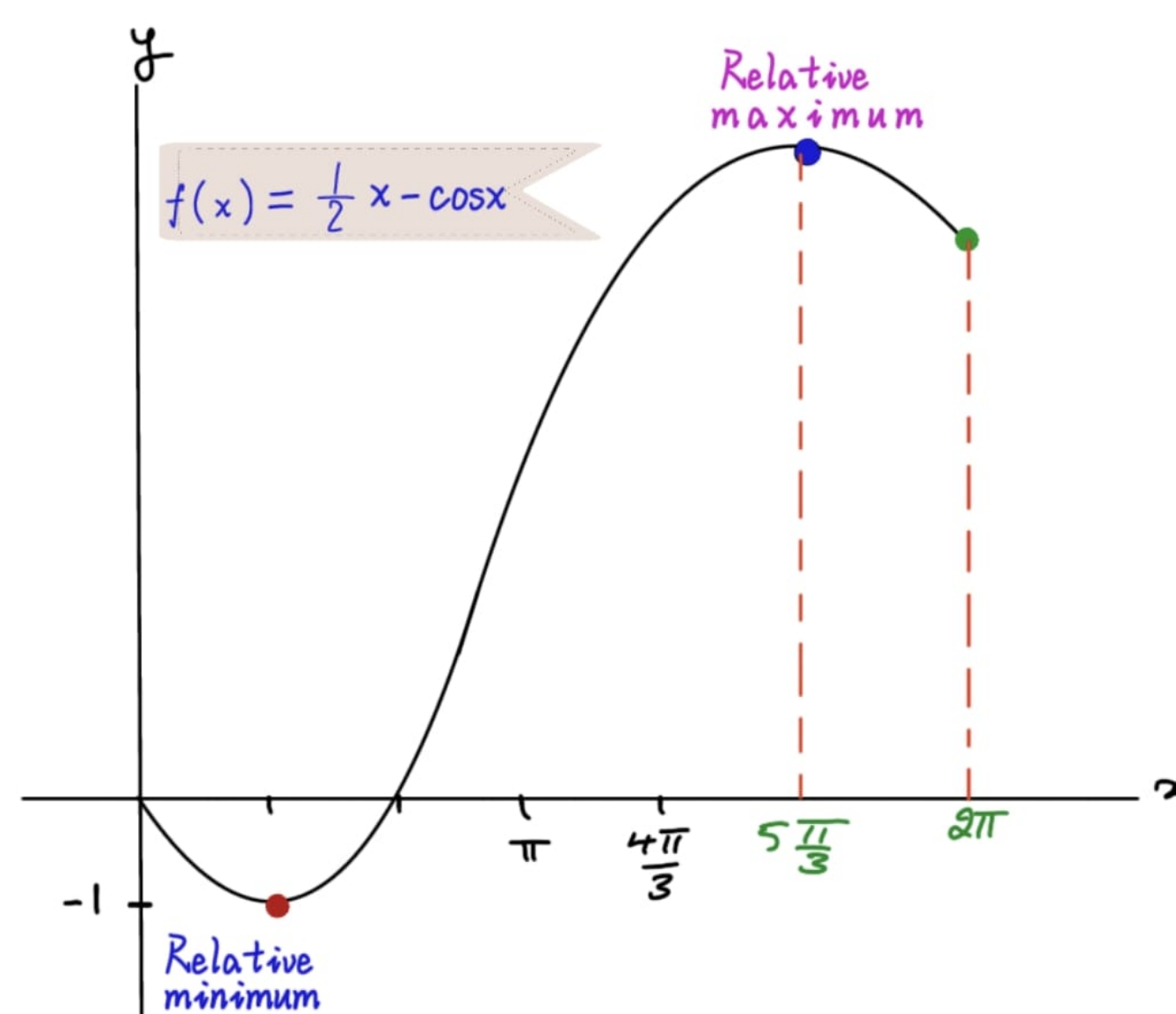
Example 1.3.9 Find the relative extrema of the function $f(x) = \frac{1}{2}x - \sin x$ in the interval $(0, 2\pi)$

Sol

To determine the critical numbers of f in this interval, set $f'(x) = 0$.

$$f'(x) = \frac{1}{2} - \cos x \Rightarrow \frac{1}{2} - \cos x = 0 \Rightarrow \cos x = \frac{1}{2} \Rightarrow x = \frac{\pi}{3}, \frac{5\pi}{3}$$

The following table summarizes the testing of the three intervals determined by these two critical



numbers.

Interval	$0 < x < \frac{\pi}{3}$	$\frac{\pi}{3} < x < \frac{5\pi}{3}$	$\frac{5\pi}{3} < x < 2\pi$
Test Value	$x = \frac{\pi}{4}$	$x = \pi$	$x = \frac{7\pi}{4}$
Sign of $f'(x)$	$f'(\frac{\pi}{4}) < 0$	$f'(\pi) > 0$	$f'(\frac{7\pi}{4}) < 0$
Conclusion	Decreasing	Increasing	Decreasing

Exercise 1.3.2 Find the local extrema of the functions:

1. $f(x) = (x^2 - 4)^{\frac{2}{3}}$

2. $f(x) = \frac{x^4 + 1}{x^2}$